Exercises: chapter 2: Integration and Administration

Exercise 1: Conceptual Understanding & Terminology

1- Define the acronym **ETL** and briefly explain the purpose of each step in the context of a Data Warehouse.
2- List and explain three distinct techniques used for **Data Extraction** as mentioned in the course
3- Explain the difference between **Full Loading** and **Incremental Loading**. When would you strictly *must* use Full Loading?
4- In the context of Data Transformation, explain **"Binning"** and provide an example of how it is used.
5- The lecture discusses **"Information Loss"** during transformation.
    a. Identify two specific transformation processes that cause information loss.
    b. Why might a Business Intelligence engineer *intentionally* choose to lose information (e.g., via Aggregation)?
6- The course material contrasts different loading plans. Explain the difference between Serial Loading and Parallel Loading. How does the technique of File Partitioning contribute to the efficiency of Parallel Loading?

Solution of Exercise 1

1. Definition of ETL

- **ETL** stands for **Extract, Transform, and Load**.
- It is an automated process that collects data from different sources, cleans and organizes it, and then loads it into a system (like a Data Warehouse) for analysis.

2. Three Data Extraction Techniques

- **OCR:** Reading text from scanned documents or images automatically.
- **APIs:** Connecting programmatically to data sources to query data.
- **Web Scraping:** collecting data from web pages by parsing HTML code.
- *(Other valid answers from course: ADC sampling, Database querying).*

3. Full Loading vs. Incremental Loading

- **Full Loading:** Loads the entire dataset from scratch. You *must* use it when no prior data exists (e.g., initializing a new warehouse).
- **Incremental Loading:** Loads only data that has changed or is new since the last load. It appends data instead of overwriting.

4. Binning (Transformation)

- **Binning** is grouping data into categories (or boxes) to facilitate analysis.
- *Example:* Grouping ages into ranges like 18-35, 36-60, etc..

5. Information Loss

- a. Causes: Lossy Data Compression (e.g., floats to integers), Filtering (removing data), or Aggregation (summarizing).
- b. Why intentional? To summarize details for high-level decision-making (e.g., seeing total monthly sales instead of every individual receipt).

6. Serial vs. Parallel Loading

- **Serial Loading:** Data is added sequentially (one after the other). It is the default plan.
- **Parallel Loading:** Data is loaded simultaneously (multiple streams) to be faster and optimized.
- **File Partitioning:** It splits a single file into smaller chunks so those chunks can be loaded simultaneously.

Exercise 2: True or False?

Determine if the following statements are True or False. If False, correct the statement.

1. The default loading plan in a Data Warehouse is **Parallel Loading**, where data is added simultaneously to speed up the process.
2. **Web Scraping** is a technique used to connect automatically to data sources via a programmatic interface to query data.
3. **Incremental Loading** is a strategy where the system deletes existing records and reloads the entire dataset from scratch every time.
4. In the **Pull Method**, the Data Warehouse subscribes to receive data, making it the best choice for real-time streaming.

Solution of Exercise 2

1. False: The default loading plan is **Serial (or sequential) loading**. Parallel loading is an optimized approach.

2. False: This describes **APIs**. Web Scraping collects data by parsing HTML from web pages.

3. False: This describes **Full Loading**. Incremental loading appends new data instead of overwriting existing records.

4. False: The **Pull Method** is useful for scheduled extractions and batch loading. For real-time streaming, the **Push Method** is used.

Exercise 3: The "Smart City" Case Study

Scenario: Imagine you are the Data Architect for a city. You need to build a Data Warehouse to analyze traffic and weather patterns. You have two main data sources:

1. **Old Traffic Logs:** 50 years of paper documents containing handwritten records of car accidents.
2. **Live IoT Sensors:** Thousands of sensors on traffic lights measuring car speed and temperature in real-time.

Questions:

- For the **Old Traffic Logs**, which specific extraction technique would you use to digitize the text?
- The city wants to protect the privacy of drivers involved in accidents mentioned in the old logs. Which transformation technique should be applied to names and license plate numbers?
- You need to load the **Live IoT Sensor** data.
    - Would you use **Push** or **Pull** methodology? Explain why.
    - Would you use **Batch Loading** or **Stream Loading**? Explain why.
- You have millions of records coming in from the sensors every hour. To speed up the process, you decide not to load the data one by one (Serial Loading).
    - What loading plan should you adopt to optimize speed?
    - The sensor system aggregates all records into **one single massive file** (50GB) every hour. Your junior engineer argues that Parallel Loading is impossible because there is only "one source file." Explain how you can still apply Parallel Loading to this single file to prove them wrong.

Solution of Exercise 3

1. Extraction for Old Logs: You would use **OCR (Optical Character Recognition)** to read text from the scanned paper documents automatically.

2. Privacy Transformation: You should use **Anonymizing or Encrypting** to hide sensitive information.

3. Loading IoT Data

- **Method: Push Method**, because the source pushes data to the warehouse and it is useful for real-time data.
- **Strategy: Stream Loading**, because it handles continuous data updates triggered by events (like sensors).

4. Plan: Parallel Loading to boost efficiency.

- We can use **File Partitioning** to split that single massive file into smaller chunks. Even though the source is just one file, these chunks can then be loaded simultaneously (parallelly) into the destination, boosting efficiency.